It's Hard to Know How Hard It Is: Mapping the Design Space of LLM Item Difficulty Estimation

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Abstract. Calibrating item difficulty traditionally requires field-testing with hundreds of students—a bottleneck as LLM-generated assessments scale. Can LLMs estimate difficulty directly from item text? We systematically compare 16 prompts across 6 models and 3 datasets (~120 configurations) and find that *how* you prompt matters more than *which* model you use. Prompts that direct the model to enumerate structural features—count prerequisites, identify error paths, list cognitive load elements—achieve $\rho = 0.65$–0.69 as a class, outperforming prompts that simulate student responses ($\rho = 0.19$–0.47 vs. baseline $\rho = 0.56$). Prompt category explains 6× more variance than model choice ($\eta^2 = 0.31$ vs. 0.05). Mechanism analysis reveals that the active ingredient is structured decomposition—breaking items into components before estimating—not the counts themselves, since buggy-rule counts carry zero difficulty signal yet the prompt achieves $\rho = 0.66$. This approach captures structural complexity visible in item text rather than population preparedness, explaining both where it succeeds and where it attenuates.

Keywords: item difficulty estimation · LLM evaluation · prompt engineering · psychometrics

1 Introduction

A well-designed test gives each student questions at the right level of challenge—hard enough to reveal what they know, easy enough to avoid frustration. Getting this right depends on knowing how difficult each question is before it reaches a student. In practice, this means field-testing items with hundreds or thousands of real students and computing how many answer correctly. In classical test theory, this proportion is called p-correct; item response theory refines it with a latent difficulty parameter b that accounts for examinee ability [9]. Either way, calibrating each item requires collecting responses from 200 to 1,000+ students depending on the IRT model [6], making difficulty estimation one of the most resource-intensive steps in assessment development.

That cost is becoming a bottleneck. LLMs can now generate assessment items at scale [17], but generated items arrive without calibrated difficulty estimates. Without such estimates, items cannot be assembled into well-targeted tests, used in adaptive systems, or compared to existing item banks. The question

is whether LLMs can also estimate difficulty—replacing or supplementing the expensive field-testing step.

The task is harder than it appears. LLMs are trained on expert-written text and tend to find most items trivial; Li, Chen et al. [14] find that high model performance paradoxically impedes difficulty estimation—models that solve items easily cannot perceive what makes them hard for humans. This "curse of knowledge" creates a systematic bias: models overestimate student ability and compress difficulty estimates toward the easy end of the scale. Expert teacher judgment, by comparison, achieves correlations of $r = 0.63$–0.66 with actual student performance [12,24]—a useful reference point, though these studies measure judgment of individual students rather than aggregate item difficulty.

Recent work shows wide variation: some studies report strong results (up to $r = 0.87$ with feature extraction and gradient boosting [19]), while others find near-zero correlations for direct estimation [1]. A systematic review of 37 papers found no consensus on methods [18], and the first shared task on difficulty prediction (BEA 2024) found best results only marginally beat baselines [26]. Li, Chen et al. [14] evaluated 20+ models and found they converge on a "machine consensus" that diverges from human perception.

A key limitation is that each study typically tests one model with one prompt on one dataset. When results disagree, it is unclear whether the discrepancy reflects model choice, prompt design, item properties, or student population. We address this by mapping the design space rather than testing any single configuration.

Our optimization target is **rank-order agreement** between LLM-predicted difficulty and empirical p-correct, measured by Spearman's ρ. We choose rank correlation because practical applications—item sequencing, adaptive test assembly, item bank stratification—depend on difficulty *ordering* rather than exact calibration. Each experimental condition prompts an LLM to estimate the proportion of students who would answer each item correctly; we then correlate these estimates against observed p-correct from real student responses.

Contributions. (1) A systematic comparison of 16 theory-grounded prompts across 6 models and 3 datasets (~120 configurations), showing that item-analysis prompts as a class outperform population-simulation prompts ($\eta^2 = 0.31$). (2) A mechanism analysis showing that directed enumeration works through structured decomposition, not counting: buggy-rule counts carry zero difficulty signal yet the prompt achieves $\rho = 0.66$. (3) A boundary condition: the method captures structural complexity visible in item text, not population preparedness—succeeding on open-ended items but attenuating on MCQs. (4) Methodological guidance on Simpson's paradox in unbalanced sweeps, minimum sample sizes ($\geq$100 items), and two-stage sequential design of experiments (DOE).

2 Related Work

We organize prior work by approach and extract testable claims from each.

Direct LLM Estimation. Razavi & Powers [19] report $r = 0.83$ (math) and $r = 0.81$ (reading) on K–5 items using GPT-4o. However, Acquaye et al. [1] find $r \approx 0$ for direct estimation on NAEP items, and Li, Chen et al. [14] find average $\rho = 0.28$ across 20+ models on four assessment domains. *Testable claim: direct estimation produces meaningful correlations ($\rho > 0.30$) on curriculum-aligned items.*

Feature Extraction + ML. Razavi & Powers [19] achieve their strongest results ($r = 0.87$) by extracting features from LLMs and training gradient-boosted models. *Testable claim: LLM-extracted features combined with ML outperform direct estimation.*

Simulated Classrooms. Acquaye et al. [1] simulate classrooms of LLM students at multiple ability levels, achieving $r = 0.75$–0.82 on NAEP items. Counter-intuitively, weaker math models predicted difficulty better than stronger ones. However, Li, Chen et al. [14] find significant misalignment between AI and human difficulty perception. *Testable claim: simulation outperforms direct estimation.*

Model Uncertainty. Zotos et al. [27] use LLM uncertainty features (first-token probability, choice-order sensitivity) to predict difficulty. *Testable claim: items the model finds uncertain are items students find difficult.*

Reasoning Augmentation. Feng et al. [10] report up to 28% MSE reduction when generating reasoning before predicting difficulty. However, Li, Jiao et al. [15] find minimal improvements from chain-of-thought prompting for fine-tuned models. *Testable claim: structured deliberation improves difficulty estimation.*

LLM-as-Respondent Approaches. A growing body of work uses LLMs to simulate student responses rather than directly estimate difficulty. Results vary widely: Liu et al. [16] report $\rho = 0.87$ calibrating IRT parameters from simulated responses on 20 college algebra items, while Srivatsa et al. [23] find that no model-prompt pair reliably simulates target grade levels across 489 NAEP items, and Sauberli et al. [21] report weak correlations ($r = 0.32$–0.56) across 18 LLMs, concluding that zero-shot LLMs should not pilot assessments. Two findings from this literature motivate our design. First, Benedetto et al. [3] show that simulation prompts engineered for one model do not transfer to others—underscoring the need for systematic cross-model evaluation. Second, Scarlatos et al. [22] achieve $r = 0.67$ by fine-tuning Llama-3.2 with DPO to create ability-aligned simulated students, demonstrating that training-based approaches can reach performance comparable to our best zero-shot results. *Testable claim: zero-shot prompt design can match simulation-based approaches without requiring training data or fine-tuning.*

Cognitive Item Models. A separate tradition predicts difficulty from structural item features. Embretson's [8] cognitive design system counts processing steps and working memory demands. Gorin [11] manipulates item features to generate

Table 1: Hypotheses tested in this study. Verdicts: ✓ = supported, × = rejected, ∼ = partial/mixed.

#	Hypothesis	Source	Prediction	Verdict
From Prior Work (Replication)				
H1	Direct estimation yields ρ>0.30	Razavi & Powers	$\rho > 0.30$	✓ .56
H2	Simulation beats direct	Acquaye et al.	$\rho_{\text{sim}} > \rho_{\text{dir}}$	× .19<.56
H3	Larger models do better	Scaling expect.	Larger → higher ρ	∼ mixed
From Learning Science Theory				
H4	Item analysis beats direct	Embretson (1998)	$\rho_{\text{ana}} > \rho_{\text{dir}}$	✓ .69>.56
H5	Prerequisite counting works	KLI (Koedinger)	More prereqs → harder	✓ .69
H6	Cog. load counting works	CLT (Sweller)	More elements → harder	✓ .67
H7	Buggy rules analysis works	Brown & Burton	More errors → harder	✓ .66
Exploratory (Design Space)				
H8	Temperature helps	Stoch. diversity	High temp → higher ρ	× η^2=.001
H9	Multi-sample averaging helps	Wisdom of crowds	$\rho_{3\text{-rep}} > \rho_1$	∼ +.03
H10	Prompt is primary lever	Design space	Prompt > model + temp	✓ open; ∼ MCQ
H11	Analysis needs capable models	Capacity limits	Analysis × capability	✓ interaction
H12	Signal transfers across data	Generalizability	Signal on D1 → D2	✓ attenuates

items at target difficulties. The KLI framework [13] posits that the number and nature of knowledge components drive learning difficulty. Recent work has used LLMs for KC tagging [17], but not to operationalize cognitive item models as difficulty estimation prompts. *Testable claim: prompts grounded in cognitive item modeling outperform atheoretical prompts.*

3 Hypotheses

We derive hypotheses from three sources: replication of prior claims, predictions from learning science theory, and exploratory questions about the design space. Table 1 lists all 12 hypotheses with their sources and predictions.

Table 2: Experimental stages.

Stage	Purpose	Dataset	Design
Screening	Map prompt space	SmartPaper (140)	15 prompts × 2–5 temps × 3 reps
Model survey	Model generalization	SmartPaper (140)	6 models × 3 prompts × 3 reps
Confirmation	Cross-dataset transfer	DBE-KT22 (168)	3 prompts × 2 temps × 3 reps
Validation	Published benchmark	BEA 2024 (595)	3 prompts × 1 temp × 3 reps

4 Method

4.1 Datasets

SmartPaper (India, Open-ended)—Primary. The SmartPaper dataset contains 140 open-ended questions from Indian state assessments across four subjects (English, Mathematics, Science, Social Science) for Grades 6–8. Each item was administered to a subset of 728,000+ students in government schools serving economically weaker sections. Ground truth is the classical p-correct (proportion scoring full marks), ranging from 0.04 to 0.83 (mean 0.29).

DBE-KT22 (South Africa, MCQ)—Confirmation. The DBE-KT22 dataset [7] contains 168 undergraduate computer science multiple-choice questions spanning 27 knowledge components, administered to 1,300+ students. Ground truth is the classical p-correct. This dataset differs from SmartPaper in item format (MCQ vs. open-ended), domain (CS vs. K–8 curriculum), and student population.

BEA 2024 Shared Task (USA, USMLE MCQ)—Validation. The BEA 2024 shared task [26] provides 595 text-only items from the United States Medical Licensing Examination (USMLE), comprising Steps 1–3 questions (667 total items minus 72 requiring images). Ground truth consists of IRT difficulty parameters from operational administrations. This benchmark allows direct comparison with other published difficulty estimation approaches.

4.2 Design Space

4.3 Prompts

All 16 prompts share a common output: the LLM predicts what proportion of students would answer each item correctly. They differ along two binary dimensions: whether the prompt directs the model to *analyze structural item features* (prerequisites, steps, cognitive demands) before estimating, and whether it directs the model to *model the student population* (reason about proficiency

Table 3: Prompt taxonomy with factor coding. Item = analyzes item structure; Pop. = models student population. All 15 prompts except synthetic_students were tested in screening.

Prompt	Item	Pop.	Theory/Prior Work
teacher			PCK (Shulman, 1986)
verbalized_sampling			Wisdom of crowds
familiarity_gradient	✓		Transfer distance
contrastive	✓		Comparative judgment
prerequisite_chain	✓		KC theory (Koedinger et al.)
cognitive_load	✓		CLT (Sweller, 1988)
cognitive_profile	✓		CLT + profiling
devil_advocate		✓	Debiasing heuristics
teacher_decomposed		✓	IRT stratification
classroom_sim		✓	Acquaye et al. (2025)
imagine_classroom		✓	Imagery-based reasoning
synthetic_students*		✓	Student simulation
error_analysis	✓	✓	Error analysis tradition
error_affordance	✓	✓	Brown & Burton (1978)
buggy_rules	✓	✓	Brown & Burton (1978)
misconception_holistic	✓	✓	Chi et al. (1994)

*Tested in model survey only ($\rho = 0.19$, worst performer).

levels, struggles). Table 3 lists all 16 prompts; 15 were tested in screening (synthetic_students was added in the model survey). All prompts are available in the project repository.

4.4 Models

Six models spanning 8B to frontier-scale: Gemini 3 Flash (screening model), GPT-4o, Llama-3.3-70B, Gemma-3-27B, Llama-4-Scout (17B active/109B MoE), and Llama-3.1-8B. Qwen3-32B was tested but excluded due to low parse rates (<50% valid responses). Table 4 reports completion and parse rates for all models; GPT-4o's 65% overall parse rate means its reported correlations are based on a non-random subset of items and may not be directly comparable.

Scope. The design space excludes few-shot prompting for three reasons. First, few-shot difficulty estimation requires exemplar items with known difficulty—contradicting the zero-shot use case we target, where items arrive without calibration data. Second, exemplar selection introduces a confound (which items, how many, in what order) that would expand the design space beyond our budget. Third, with long item texts and rubrics, few-shot prompts approach context limits on smaller models and increase per-item cost. Few-shot prompting is a natural extension for settings where calibrated exemplars are available.

Table 4: Model survey parse rates (aggregated across all prompt framings). Screening (Gemini only) achieved 100% across all 14,700 responses.

Model	Expected	Present	Parsed	File%	Parse%
Gemini 3 Flash	2,100	2,100	2,100	100.0	100.0
GPT-4o	2,100	1,400	915	66.7	65.4
Llama 3.3 70B	2,100	1,949	1,948	92.8	99.9
Llama 4 Scout	2,100	1,955	1,952	93.1	99.8
Gemma 3 27B	2,100	1,502	1,502	71.5	100.0
Llama 3.1 8B	2,100	1,948	1,918	92.8	98.5
Qwen 3 32B (excl.)	1,260	472	426	37.5	90.3

4.5 Statistical Approach

Our primary metric is Spearman's ρ between LLM-predicted and observed p-correct. We report 95% bootstrap confidence intervals (10,000 resamples) for primary results and mean absolute error (MAE) as a calibration measure. Each condition uses 3 replications; reported ρ is the averaged-prediction correlation (mean prediction across reps correlated with ground truth). Correlations are computed only over successfully parsed items; unparseable responses are treated as missing data.

5 Results

We present results in six parts: the screening results, the group-level item-analysis finding, factors with minimal effects, cross-dataset generalization, practical implications, and statistical caveats on confidence intervals and replication noise.

5.1 Item Analysis Prompts Achieve $\rho = 0.65$–0.69

Fifteen prompts were screened on SmartPaper (140 open-ended items) using Gemini 3 Flash at 2–5 temperature settings with 3 replications per condition. The top three prompts—all directing the model to analyze structural item features before estimating difficulty—achieved $\rho = 0.65$–0.69 (Table 5, Figure 1): **prerequisite_chain** ($\rho = 0.686$) enumerates knowledge prerequisites before estimating; **cognitive_load** ($\rho = 0.673$) counts interacting memory elements; and **buggy_rules** ($\rho = 0.655$) identifies procedural error paths. For context, meta-analyses report that experienced teachers can predict individual student performance with $r = 0.63$–0.66 [12,24]. The constructs differ: teachers judge individual students while we predict aggregate item difficulty. Direct comparison is inappropriate, but the magnitude suggests LLM estimates capture meaningful difficulty information.

Table 5: Prompt screening results (SmartPaper, $n = 140$): top 10 of 15 screened prompts, ranked by best ρ. Five additional prompts (verbalized_sampling, familiarity_gradient, imagine_classroom, teacher_decomposed, error_affordance) achieved $\rho < 0.55$. CIs shown for prompts advanced to model survey.

Prompt	Item	Pop.	Best ρ	Temp	95% CI	MAE
prerequisite_chain	✓		**0.686**	0.5	[.576,.775]	.155
cognitive_load	✓		**0.673**	2.0	[.550,.766]	.190
buggy_rules	✓	✓	**0.655**	1.0	[.532,.752]	.117
misconception_holistic	✓	✓	0.636	2.0	—	.204
error_analysis	✓	✓	0.596	2.0	—	.121
devil_advocate		✓	0.596	1.0	—	.098
cognitive_profile	✓		0.586	1.0	[.456,.693]	.214
contrastive	✓		0.584	1.0	—	.123
classroom_sim		✓	0.562	2.0	—	.240
teacher			0.555	1.0	[.422,.664]	.439

5.2 Item Analysis Outperforms Population Simulation

Grouping prompts by their 2×2 factor membership (Table 3) reveals a clear pattern: item analysis is the primary driver. Prompts that model student populations without analyzing item structure perform *worse* than baseline. Mean ρ by group: item analysis only = 0.61 (5 prompts); item + population = 0.59 (4 prompts); neither factor = 0.52 (2 prompts); population only = 0.47 (4 prompts). This parallels Embretson's [8] cognitive design system, where item difficulty is predicted from counts of processing steps and working memory demands. The LLM operationalizes similar constructs when directed to enumerate prerequisites or interacting elements.

The critical distinction is *item-centric* vs. *population-centric* framing. Hybrid prompts like buggy_rules ($\rho = 0.66$) succeed because "identify procedural errors students might make" is fundamentally item analysis—it asks what error paths the item affords, not how a simulated population would respond. Population modeling *alone* is unreliable: explicit student simulation (synthetic_students: $\rho = 0.19$) performs worst, consistent with Li, Chen et al.'s [14] finding that proficiency simulation does not reliably improve difficulty estimation. Framing analysis through student behavior works; simulating student populations does not.

5.3 Factors with Minimal Effects

Three factors each explained <5% of variance (unique $\eta^2 < 0.05$):

Temperature. Most prompts show $\Delta\rho < 0.05$ across temperatures 0.5–2.0. The exception: prerequisite_chain peaks at temperature 0.5 while cognitive_load peaks at 2.0.

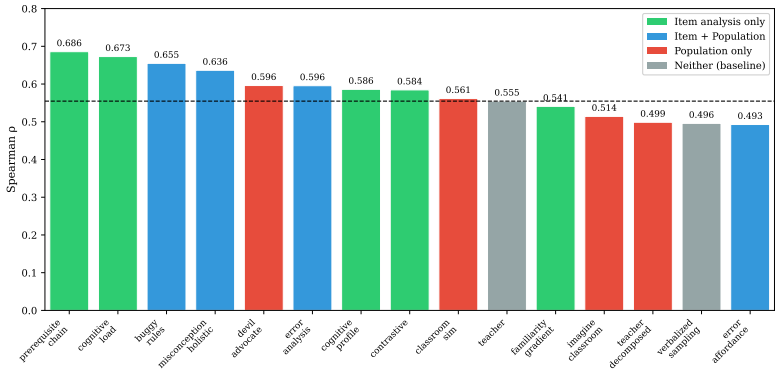


Fig. 1: Prompt screening results on SmartPaper ($n = 140$). Color indicates factor membership: green = item analysis only, blue = item + population, red = population only, grey = neither. Dashed line = teacher-role prompt baseline. Error bars show 95% bootstrap CIs.

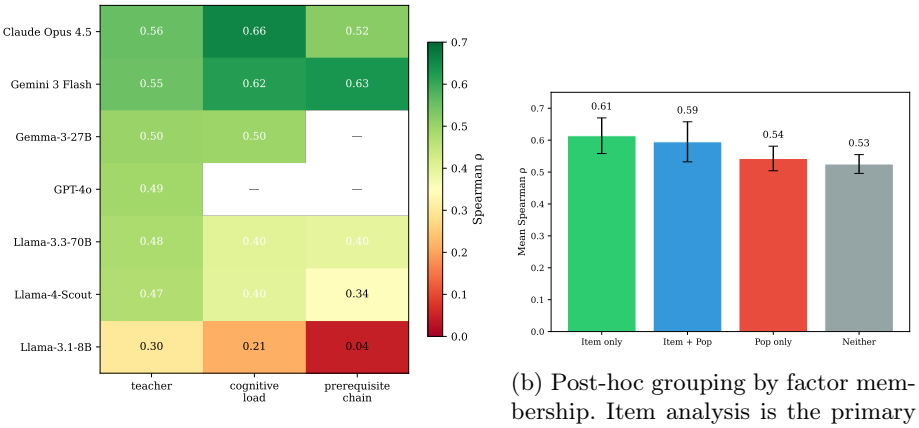
Table 6: Model survey (ρ, SmartPaper, $t = 1.0$). "—" indicates condition not run or <50 valid responses.

Model	Params	teacher	cog_load	prereq
Gemini 3 Flash	—	0.55	0.62	**0.66**
Gemma-3-27B	27B	0.50	0.50	—
GPT-4o	—	0.49	—	—
Llama-3.3-70B	70B	0.48	0.40	0.40
Llama-4-Scout	17B/109B	0.47	0.40	0.35
Llama-3.1-8B	8B	0.30	0.21	0.04

Model size. Among frontier models, size does not predict success. GPT-4o ($\rho = 0.49$) underperforms Gemini 3 Flash ($\rho = 0.55$) and matches the smaller Gemma-3-27B. A capacity floor exists—Llama-3.1-8B struggles ($\rho = 0.30$)—but above that threshold, other factors dominate (Table 6).

Deliberation. Undirected chain-of-thought reasoning degrades performance. The structured prompts that work use *directed* analysis—they specify exactly what to enumerate (prerequisites, interacting elements, error paths). Undirected "think step by step" produces unfocused reasoning that drifts toward solving the item.

Surface features. Text length correlates $\rho = -0.44$ with difficulty on SmartPaper (shorter questions are harder). Controlling for text length, partial $\rho = 0.59$–0.66 for item analysis prompts—the correlation changes minimally ($\Delta\rho \leq 0.06$), confirming that LLM predictions are not merely proxies for question length.



(a) Model survey. Frontier models achieve $\rho > 0.50$; Llama-8B fails on structured prompts.

(b) Post-hoc grouping by factor membership. Item analysis is the primary driver; population modeling alone underperforms baseline.

Fig. 2: (a) Model × prompt heatmap on SmartPaper. (b) Mean ρ by 2×2 factor membership with 95% CIs.

Table 7: Cross-dataset generalization (Gemini 3 Flash). SmartPaper and DBE-KT22 use oracle temperature (best per dataset); BEA uses $t = 1.0$. Oracle selection requires ground truth and represents an upper bound, not a practical recommendation.

Dataset	n	teacher	prereq	buggy	Best	95% CI
SmartPaper (open-ended)	140	0.56	**0.69**	0.66		[.58,.78]
DBE-KT22 (CS MCQ)	168	0.52	0.53	**0.58**		[.46,.68]
BEA 2024 (USMLE MCQ)	595	0.45	0.44	**0.45**		[.39,.52]

Model × prompt interaction. Item analysis prompts help capable models but hurt weak ones. On Gemini, prerequisite_chain gains 0.08 over teacher; on Llama-8B, it loses 0.26.

5.4 Generalization Across Datasets

We validated on two additional datasets: DBE-KT22 (168 undergraduate CS MCQs) and the BEA 2024 shared task (595 USMLE medical MCQs with IRT difficulty). Table 7 shows results.

The signal transfers: all prompts achieve significant correlations on all three datasets. However, the item analysis advantage attenuates on MCQs. On SmartPaper, prerequisite_chain beats teacher by 0.13; on DBE-KT22, buggy_rules achieves the highest $\rho = 0.58$ but the gap over teacher is only 0.06; on BEA, all three prompts are equivalent ($\rho \approx 0.45$).

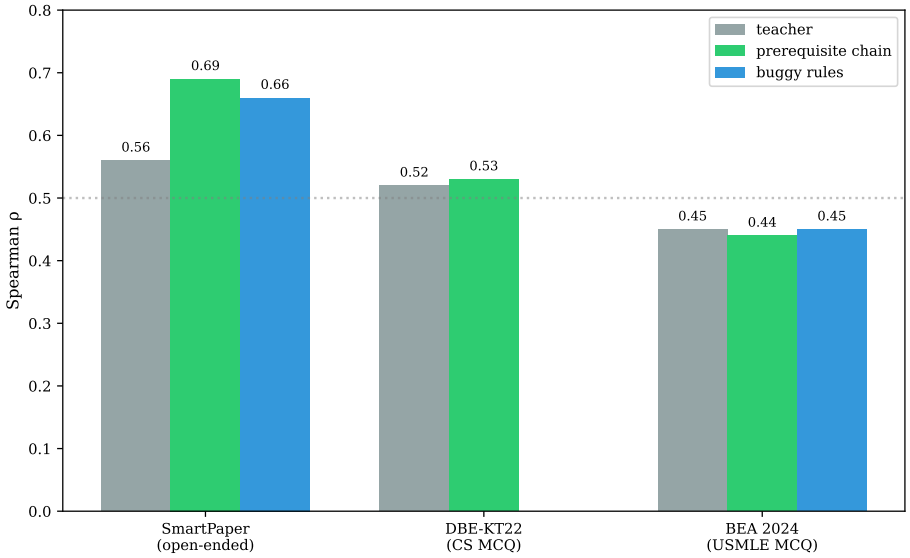


Fig. 3: Cross-dataset generalization. The item-analysis advantage (prerequisite_chain, green) is strongest on open-ended items and attenuates on MCQs, where the teacher-role prompt (grey) nearly matches structured prompts.

This suggests a **structural predictability hypothesis**: LLMs estimate difficulty well when it varies across knowledge domains (open-ended items spanning topics), but the advantage of counting prompts diminishes when difficulty is driven by distractor design within a domain. The teacher-role prompt is equally effective for MCQs.

BEA 2024 Benchmark Comparison. The BEA 2024 shared task [26] evaluated difficulty prediction on 667 USMLE items using RMSE on IRT difficulty as the primary metric. The winning system (EduTec) achieved RMSE=0.299, barely improving over a mean-prediction baseline of 0.311; the best post-competition result (UnibucLLM, support vector regression with BERT features [20]) achieved RMSE=0.281 with Kendall $\tau = 0.28$. Our zero-shot predictions show systematic bias: models overestimate student ability. However, when calibrated via linear scaling on the BEA training set, our three-prompt ensemble (averaging teacher, prerequisite_chain, buggy_rules) achieves RMSE=0.280 and $\tau = 0.31$. This matches or slightly exceeds the best published result—despite using prompts designed for K–8 education rather than medical licensing exams. This suggests that zero-shot rank-order signal, combined with simple calibration, can match supervised feature-extraction approaches.

5.5 Practical Implications

At approximately \$0.01 per item (3 replications × Gemini 3 Flash at \$0.50/\$3.00 per 1M tokens input/output), LLM difficulty estimation is orders of magnitude cheaper than field testing. Rankings transfer across datasets; absolute calibration does not—models overestimate student ability on SmartPaper. Notably, item analysis prompts achieve lower MAE (0.12–0.19) than the teacher-role prompt (MAE=0.44), but the rank-order is preserved regardless of calibration bias.

For practitioners: use a capable model (Gemini, GPT-4o), prompt it to analyze structural item features, and average 3+ replications. For standard MCQs, the teacher-role prompt suffices.

5.6 Statistical Caveats

Two caveats qualify the results above.

Overlapping confidence intervals. With 140 items, the 95% CI for $\rho = 0.69$ spans [.58, .78]—a width of 0.20. Adjacent prompts in Table 5 are *not* reliably distinguishable: the top-ranked prerequisite_chain overlaps substantially with the tenth-ranked teacher. Our primary claims are therefore at the **group level**: item-analysis prompts as a class ($\bar{\rho} = 0.61$, 5 prompts) outperform population-only prompts ($\bar{\rho} = 0.47$, 4 prompts). The ANOVA effect size for this grouping ($\eta^2 = 0.31$) is robust; individual prompt rankings within a group are suggestive, not definitive. Distinguishing individual prompts at $p < .05$ would require ∼400+ items—a scale achieved only on BEA, where all three prompts converged to $\rho \approx 0.45$.

Replication noise. Pairwise Spearman correlations between the three independent replications averaged $\bar{\rho} = 0.84$ (range 0.72–0.98), but stability varied with prompt type. The teacher prompt was highly stable ($\rho_{\text{inter}} = 0.97$; 7–11% of items shifted >20 rank positions). Structured prompts were substantially noisier: cognitive_load showed $\rho_{\text{inter}} = 0.72$–$0.78$, with up to 50% of items shifting >20 positions between runs. This presents a tradeoff: the teacher prompt produces consistent but less accurate estimates; structured prompts produce more accurate but noisier estimates that *require* multi-replication averaging. Averaging three replications of prerequisite_chain raised ρ from 0.63–0.67 (single run) to 0.69.

6 What Explains the Enumeration Advantage?

The screening results show *that* directed enumeration prompts outperform baselines. This section asks *why*, analyzing the raw text outputs from prerequisite_chain, cognitive_load, and buggy_rules to determine what role the enumerated features play in the model's estimates.

Table 8: Correlations between enumerated counts, model predictions, and ground truth p-correct (SmartPaper, $n = 140$, Gemini 3 Flash). All values are Spearman's ρ with 95% bootstrap CIs.

	Count → GT	Prediction → GT	Count → Prediction
prerequisite_chain	−0.36 [−.50, −.20]	+0.69 [+.59, +.77]	−0.48 [−.61, −.32]
cognitive_load	−0.25 [−.40, −.09]	+0.67 [+.56, +.78]	−0.33 [−.48, −.16]
buggy_rules (total)	−0.01 [−.17, +.16]	+0.66 [+.54, +.75]	−0.01 [−.18, +.16]
teacher (baseline)	—	+0.55 [+.40, +.67]	—

Table 9: Partial correlations: does the model add information beyond the count?

Prompt	r(pred, GT | count)	r(count, GT | pred)
prerequisite_chain	+0.63	−0.05
cognitive_load	+0.65	−0.04
buggy_rules	+0.66	+0.00

6.1 Do Enumerated Counts Carry Difficulty Signal?

Each prerequisite_chain output contains an explicit `COUNT: N` field. We extracted counts across all 140 SmartPaper items (averaged over 3 replications) and correlated them with observed p-correct (Table 8).

Prerequisite counts weakly predict difficulty ($\rho = -0.36$): items with more identified prerequisites are harder. Cognitive load counts show a weaker signal ($\rho = -0.25$). But buggy rule counts show *no* relationship with difficulty ($\rho = -0.01$)—despite buggy_rules achieving $\rho = 0.66$ overall.

This dissociation complicates a simple "counting features predicts difficulty" story. For prerequisite_chain, the enumeration produces a byproduct (a count) that correlates with difficulty. For buggy_rules, the enumeration improves the estimate while the count itself carries no signal. The common factor across all three prompts is the *process* of structured decomposition, not the *product*.

6.2 The Count Is a Byproduct, Not the Mechanism

If models were mapping counts to estimates, controlling for the count should eliminate the prediction–ground-truth correlation. We tested this with partial correlations (Table 9).

Controlling for count, the prediction retains nearly all its power ($r = 0.63$–0.66). Controlling for the prediction, the count adds nothing ($r \approx 0$). Whatever information the count captures is entirely subsumed by the model's holistic estimate.

Two interpretations are consistent with this evidence. First, the model may integrate the *content* of the enumerated features—not just how many prerequisites, but which ones, and how they interact. Second, the decomposition may

Table 10: Mean p-correct by prerequisite count bin (SmartPaper, Gemini 3 Flash).

Count bin	n items	Mean p-correct	SD
4–5 prerequisites	9	0.53	0.10
6–7 prerequisites	107	0.28	0.16
8+ prerequisites	24	0.26	0.12

Table 11: Prerequisite count characteristics by model. Models that produce meaningful counts (negative count→GT correlation) also produce better final predictions.

Model	Mean count	SD	Count → GT ρ	Pred → GT ρ
Gemini 3 Flash	6.8	0.8	−0.39	+0.66
Llama-3.3-70B	6.9	1.2	−0.46	+0.40
Llama-4-Scout	7.1	1.3	−0.39	+0.35
Llama-3.1-8B	4.3	1.9	+0.03	−0.04

simply shift the model into an analytical processing mode that produces better estimates regardless of what specifically gets enumerated. We cannot distinguish these from the current data. What we can say is that the active ingredient is the structured decomposition process—instructing the model to break items into components before estimating—rather than any particular count or feature.

6.3 The Count–Difficulty Relationship Is Monotonic

Despite the count not being the mechanism, it tracks difficulty in the expected direction. Binning items by prerequisite count (Table 10):

Items with 4–5 prerequisites are roughly twice as easy as items with 6+. But the relationship compresses above 7 prerequisites: additional complexity does not linearly increase difficulty. The same pattern holds for cognitive load elements (4–5: $p = 0.37$; 6–7: $p = 0.29$; 8+: $p = 0.26$). This compression is consistent with floor effects in item difficulty—once an item demands enough independent competencies, most students in the population fail regardless.

Note that 107 of 140 items (76%) fall in the 6–7 prerequisite bin, suggesting the model has limited count resolution. The prediction's superiority over the count likely reflects finer-grained distinctions within this large bin—distinctions captured by the content of the decomposition rather than the count.

6.4 Enumeration Quality Depends on Model Capability

The cross-model comparison reveals that not all models produce useful decompositions (Table 11).

Capable models produce 6–7 prerequisites per item with low variance and counts that correlate $\rho = -0.39$ to -0.46 with difficulty. Llama-3.1-8B produces fewer, noisier counts (mean 4.3, SD 1.9) with zero count–difficulty correlation, and its final predictions also fail entirely ($\rho = -0.04$). This explains the model×prompt interaction from Section **??**: structured enumeration requires a model capable of generating pedagogically meaningful decompositions. Below that capability threshold, the decomposition produces noise rather than signal, and the scaffold collapses.

An asymmetry in Table 11 deserves comment: Llama-70B and Scout produce counts with *stronger* difficulty correlations than Gemini ($\rho = -0.46$ vs. -0.39) yet achieve *weaker* final predictions ($\rho = 0.40$ and 0.35 vs. 0.66). Generating a good decomposition is necessary but not sufficient—the model must also integrate the decomposition into an accurate estimate, and this integration step is where Gemini excels.

6.5 Qualitative Examination

We examined Gemini's prerequisite lists for a stratified sample of items. These examples are illustrative, not a systematic validity study—that would require domain expert ratings, which we leave to future work. Three cases:

Easy item ($p = 0.59$, 6 prerequisites). For a Grade 7 English comprehension question ("Why did the king announce a reward?"), the model identifies: reading comprehension, understanding "why" questions, vocabulary ("announce," "reward"), text scanning, information retrieval, basic sentence construction. The list is concrete and appropriate for the target population.

Hard item ($p = 0.05$, 7 prerequisites). For a Grade 8 English question asking students to identify the theme of a poem, the model adds prerequisites absent from the easy item: *abstract reasoning* from literal narrative to symbolic meaning, the concept of "theme" as distinct from summary, *synthesis* across four stanzas, and cultural bridging between philosophical concepts and lived experience. Whether these specific prerequisites are the *right* ones is debatable, but the model correctly identifies that this item demands qualitatively different cognitive operations.

Instructive failure ($p = 0.40$, 9 prerequisites). For "Solve $4x + 2 = 10$" (Grade 7), the model lists 9 prerequisites: reading comprehension, variable concept, inverse operations (subtraction and division), integer arithmetic, and multi-step reasoning. The count overestimates difficulty because the model decomposes the item into atomic cognitive steps appropriate for a naïve learner, ignoring that students at this curricular stage have typically practiced linear equations to automaticity. This reveals the method's fundamental limitation: **it estimates difficulty from structural complexity visible in the item text, not from the population's preparedness**. These diverge whenever students have practiced a procedure to fluency, or conversely, when an item's difficulty stems from factors

invisible in the text (prior exposure, test anxiety, language barriers beyond what the model can infer).

7 Discussion

The Role of Structured Decomposition. Section 6 shows that directed enumeration works through the *process* of structured decomposition rather than the product of counting. The mediation analysis (Table 9) demonstrates that the model's estimate subsumes any information in the count, and the buggy_rules dissociation—zero count signal yet strong predictions—confirms that the active ingredient is decomposition itself. Two interpretations remain consistent with the evidence: decomposition may activate relevant pedagogical knowledge that otherwise stays latent, or it may simply prevent the model from drifting toward solving the item (as undirected chain-of-thought does). Distinguishing these would require ablation studies that preserve the decomposition structure while randomizing the content—a direction for future work.

Boundary Conditions. The cross-dataset results (Section 5.4) and the instructive failure in the mechanism analysis (Section 6.5) converge on the same boundary: LLM estimation succeeds when difficulty is driven by structural features visible in the item text, and fails when it is driven by population-specific factors (prior exposure, procedural fluency, distractor plausibility) that leave no textual trace. Practical heuristic: use structured prompts for open-ended items spanning knowledge domains; use the teacher-role prompt for MCQs within a domain.

Methodological Recommendations. Our design-space exploration surfaced three lessons. (1) *Beware Simpson's paradox in unbalanced designs.* We initially swept temperature only for promising prompts; a marginal ANOVA showed temperature as dominant ($\eta^2 = 0.74$), but this was an artifact—low temperatures were confounded with high-performing prompts. When restricted to balanced conditions, temperature explained $< 1\%$ of variance. (2) *Use two-stage sequential DOE.* Screen all prompts cheaply (1 rep, single temp) to eliminate the bottom half, then run a balanced factorial on survivors. (3) *Validate on $\geq$100 items.* Our 20-item probe yielded $\rho = 0.46$; the full 140-item set achieved $\rho = 0.55$. Bootstrap analysis showed only 2.5% of 20-item subsets would produce $\rho \geq 0.50$. Many published results use <50 items—treat such samples as exploratory only.

Limitations. Six limitations warrant mention. First, all 15 prompts were screened on a single model (Gemini 3 Flash); the model survey shows effects generalize but with diminished magnitude. Second, with 140 items, individual prompt rankings have wide confidence intervals (Table 5); our group-level claims are robust but specific prompt comparisons are not. Third, calibration does not transfer—models overestimate ability on SmartPaper but rankings are preserved, so population-specific adjustment would be needed for absolute predictions. Fourth, structured prompts require multi-replication averaging due to substantial inter-run noise (up to 50% of items shifting >20 rank positions); a single

run is unreliable. Fifth, we do not compare to fine-tuned models; fine-tuned BERT/RoBERTa can outperform zero-shot LLMs when training data is available [15], but our approach targets the zero-shot case where items arrive without calibration data. Sixth, and most fundamentally: as the mechanism analysis (Section 6.5) illustrates, LLM estimation captures *structural complexity visible in the item text*, not *population preparedness*. The model decomposes a linear equation into 9 atomic prerequisites because it cannot know that students have practiced this procedure to automaticity. Field testing remains essential for calibrating difficulty to specific populations.

Future Work. Three directions warrant investigation. First, batching multiple items per prompt showed preliminary improvements (+0.03–0.21 ρ) but requires systematic validation; if robust, this would further reduce per-item costs. Second, extended reasoning models (o1, DeepSeek-R1) may behave differently than the standard models tested here—our preliminary finding that thinking tokens *hurt* performance deserves replication on reasoning-optimized architectures. Third, a systematic qualitative validation study—having domain experts rate the pedagogical accuracy of LLM-generated prerequisite lists—would strengthen the evidence that decomposition activates genuine content knowledge rather than plausible-sounding boilerplate. An ablation preserving the decomposition format while randomizing content could isolate whether the structure or the substance drives the improvement.

8 Conclusion

We mapped the design space for LLM item difficulty estimation across 16 prompts, 6 models, and 3 datasets (~120 configurations, ~$100 in API calls). Item-analysis prompts as a class ($\bar{\rho} = 0.61$) outperform population-simulation prompts ($\bar{\rho} = 0.47$), though individual prompt rankings are not reliably distinguishable with 140 items. Mechanism analysis reveals that enumeration works through structured decomposition—the model integrates the content of enumerated features, not just counts, and buggy-rule enumeration improves estimates despite counts carrying no difficulty signal. This advantage attenuates on MCQs ($\rho \approx 0.45$–0.58), where difficulty is driven by distractor design rather than structural complexity. At ~$0.01 per item, LLM estimates can triage items into difficulty bands, but structured prompts require multi-replication averaging and absolute predictions require population-specific calibration.

Ethics Statement. This study uses anonymized, aggregated student response data. SmartPaper data consists of item-level statistics without individual student identifiers. DBE-KT22 and BEA 2024 are publicly available research datasets. No personally identifiable information was accessed or processed.

Data and Code Availability. Prompts and analysis code are available in the project repository. SmartPaper data are available upon request from the orig-

inating organization. DBE-KT22 is publicly available [7]. BEA 2024 data are available through the shared task organizers.

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A Example Prompts

In the prompts below, `{{...}}` denotes template variables filled per item before sending (e.g., `{{subject}}`, `{{question_text}}`). Capitalized labels like `PREREQUISITES:` are output format instructions for the model.

Teacher (baseline).

```
You are an experienced teacher in {{subject}} for Grade {{N}} students
in India.
For this open-ended question, estimate what proportion of students would
score full marks.
Question: {{question_text}}
Rubric: {{rubric}}
Maximum score: {{max_score}}
Think about:
- What specific errors or misunderstandings would cause students to
lose marks?
- How clearly does the question communicate what's expected?
- What prerequisite knowledge is needed?
- How likely are students at this grade level to have that knowledge?
Respond with ONLY a number between 0 and 1 representing the proportion
of students who would get full marks. For example: 0.45
```

Prerequisite Chain (best performer, $\rho = 0.69$).

```
{{population_context}}
For this question, identify the prerequisite knowledge and skills a
student needs. Count how many independent things must ALL go right for
a correct answer. Each prerequisite is a potential failure point.
```

```
Examples of prerequisites: reading comprehension, specific vocabulary,
a math operation, a concept definition, multi-step reasoning, writing
ability.
{{question, rubric, max_score}}
List the prerequisites, then estimate what proportion would get full
marks.
PREREQUISITES:
COUNT:
Respond with ONLY a number between 0 and 1 on the last line.
```

Buggy Rules (best hybrid, $\rho = 0.66$).

```
You are an expert in mathematical cognition and systematic student errors
(Brown & Burton, 1978).
{{population_context}}
For the following test item, analyze the cognitive demands:
{{grade, subject, question, rubric, max_score}}
Step 1: List the specific procedural steps a student must execute correctly.
Step 2: For each step, identify any known ‘‘buggy rules’’ -- systematic
procedural errors students commonly make (e.g., subtracting smaller
from larger regardless of position, forgetting to carry).
Step 3: Consider the target student population.
Step 4: Taking into account ALL of the above analysis holistically,
estimate what proportion of students would produce the fully correct
answer.
Respond with ONLY a number between 0 and 1 on the last line.
```

Synthetic Students (worst performer, $\rho = 0.19$). This two-stage approach first generates 10 student personas, then simulates each attempting each item.

Stage 1 — Persona generation:

```
Generate 10 diverse student profiles for a Class {{N}} government school
in India. {{population_context}}
The class distribution should reflect a typical government school:
- 4 students: Below Basic (barely literate, struggle with basic concepts)
- 3 students: Basic (can read Hindi well, weak English)
- 2 students: Competent (understand most concepts, some errors)
- 1 student: Advanced (strong understanding, rarely makes mistakes)
STUDENT 1: ... | Level: ... | ...
...
```

Stage 2 — Student simulation (per item, per persona):

```
You are role-playing as this student: {{persona}}
{{grade, question, rubric, max_score}}
Role-play this specific student attempting this question. Consider their
reading ability, knowledge level, attention, and typical behaviors.
Write their actual response as they would write it, then score it.
RESPONSE:
SCORE:
```